1)

$Y = 2\sqrt{u}$, $u \sim Unif[0,1]$

a) $u = 15$ -> $Y = 2\sqrt{0,15} = 0,77$

b) $P(Y \leq 0,6) = F_X^{-1}(0,6)$ (kontinuerlig)

$Y^{-1} = \left(\frac{1}{2}x\right)^2 = \frac{1}{4}x^2$

$P(Y \leq 0,6) = \frac{1}{4}x^2 \Big|_0^{0,6} = 0,09$

$Y = F_Y(u)$? (i denne oppg: tydelig)

$P(Y \leq 0,6) = 1 - P(Y > 0,6)$

$F^{-1}(0,6) = 1 - F(0,6)$

2) a) $f(x) = \frac{8}{x^3} \Rightarrow F_X(x) = \int f(x)\,dx$

$$= 8\int \frac{1}{x^3}\,dx = 8 \cdot \left[\frac{1}{(-2)} x^{-2}\right] + C$$

$$\int_2^\infty f(x)\,dx = 1 \Rightarrow \left(\frac{-4}{\infty} + C\right) - \left(\frac{-4}{4} + C\right) = 1$$

$$C - 1 + C = 1$$

$$C = 1$$

$$F_X(x) = 1 - \frac{4}{x^2} = \frac{x^2 - 4}{x^2}$$

b) $\int_2^x \frac{8}{z^3}\,dz = 0{,}95$

$$1 - \frac{4}{z^2}\Big|_2^x = 0{,}95$$

$$1 - \frac{4}{x^2} - \left(1 - \frac{4}{4}\right) = 0{,}95$$

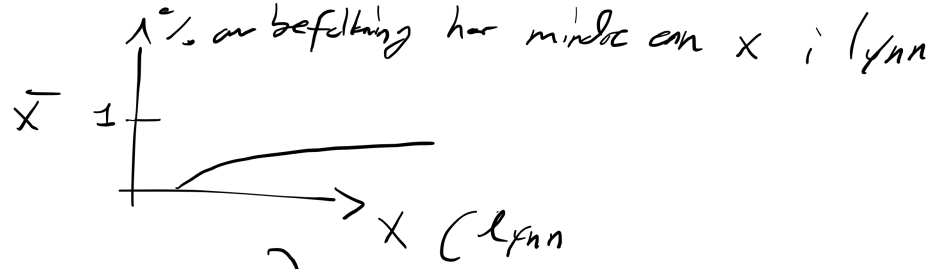
$F_X(x)$ = # personer med lykkes mellom 200 000 og $x \cdot (100\,000)$

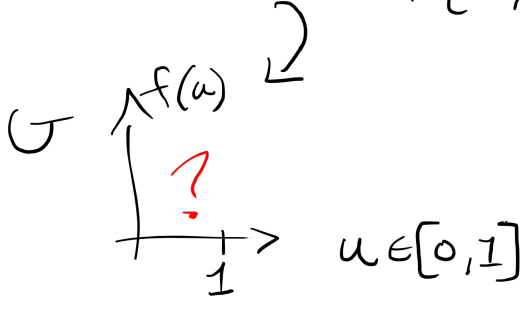
$(= 0{,}95\%)$

$\frac{4}{x^2} = 0{,}05 \Rightarrow x = 8{,}944$ | = 894 400 kr

c)

$U \leftrightarrow \bar{X}$, $Bild_{\bar{X}}(\bar{x}) = [0,1]$





$f(u) = ?$

$\bar{x} = \bar{F}^{-1}(u)$

$\rightarrow u = F(x)$

$f(u) = \text{Invecs}\left(1 - \frac{4}{x^2}\right)$

$1 - \frac{4}{u^2} = x$

$1 - x = \frac{4}{u^2}$

$\int \rightarrow \frac{4}{1-x} = u^2$

$u = \sqrt{\frac{4}{1-x}}$

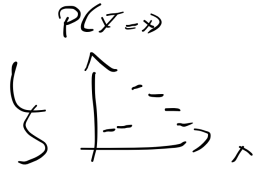
$$f(\omega) = \sqrt{\frac{4}{1-\omega}}$$

3) X = # solgte motorer per måned

X = Diskret stokastisk variabel

a) $E[X] = ?$

x:	0	1	2	3	4	5
$P(X=x)$:	0,1	0,1	0,3	0,25	0,15	0,1



$$E[X] = \sum_x x \cdot P(X=x) = 0 \cdot 0{,}1 + 1 \cdot 0{,}1 + 2 \cdot 0{,}3 + 3 \cdot 0{,}25 + \dots$$

$$= 2{,}550$$

$$Var[\bar{X}] = E\left[(\bar{X} - E[\bar{X}])^2\right] = E\left[(\bar{X} - 2{,}55)^2\right]$$

$$= E[\bar{X}^2] - (E[\bar{X}])^2 = E[\bar{X}^2] - 2{,}55^2$$

$$\begin{aligned} E[\bar{X}^2] = {} & 0^2 \cdot 0{,}1 && 0 \\ & + 1^2 \cdot 0{,}11 && + 0{,}11 \\ & + 2^2 \cdot 0{,}3 && + 1{,}2 \\ & + 3^2 \cdot 0{,}25 && + 2{,}25 \\ & + 4^2 \cdot 0{,}15 && + 1{,}2 \\ & + 5^2 \cdot 0{,}1 && + 2{,}5 \end{aligned} = \begin{aligned} & 1{,}3 \\ & + 2{,}25 \\ & + 3{,}7 \end{aligned} = 7{,}25$$

$$2{,}55^2 = 6{,}5025$$

$$Var[\bar{X}] \approx 1{,}947$$

4) T = forsinkelse [min] | kontinuerlig stok.

T

$$f(t) = \begin{cases} 1{,}5e^{-1{,}5t}, & t > 0 \\ 0, & \text{ellers} \end{cases}$$

$$P(T > 0{,}5) = 1 - P(T < 0{,}5)$$

$$= 1 - F_T^{-1}(0{,}5)$$

$$= 1 - \text{Invers}\left[\int_0^{0{,}5} f(t)\,dt\right]$$

$$F(t) = \int f(t)\,dt = -e^{-1{,}5t}$$

$$-y = e^{-1{,}5t}$$

$$F_T^{-1}(t): \quad \ln -y = -1{,}5t$$

$$\rightarrow t = \frac{\ln|y|}{1{,}5}$$

1. Integrer ($\text{Finn } F_T(t)$)
2. Finn invers
3. Invers $\Big|_0^{0{,}5}$

Men siden jeg skal ha $P(T > 0{,}5)$

kan jeg $1 - F_T(0{,}5)$?

$$-e^{-1{,}5t}\Big|_0^{0{,}5} = -e^{-0{,}75} + 1 = 0{,}5276$$

$1 - 0{,}5276 = 0{,}4472$ | Riktig

$$P(T > 0{,}5) = 1 - P(T < 0{,}5)$$

$$F_T(0{,}5) = 1 - F_T^{-1}(0{,}5)$$

$$= 1 - T$$

$T = P(x > 0{,}5)$?

$= F_T^{-1}(0{,}5)$

b) $$P(\bar{X} > 3 \mid \bar{X} > 0{,}5) = \frac{P(\bar{X} > 3 \cap \bar{X} > 0{,}5)}{P(0{,}5)}$$

$$= \frac{P(0{,}5 \mid 3)\, P(3)}{P(0{,}5)}$$

$$P(X>3 \mid X>0{,}5) = \frac{P(3)}{P(0{,}5)} \quad \left(P(0{,}5 \mid 3) = 100\%\right)$$

$$= \frac{e^{-1{,}5 \cdot 3}}{e^{-1{,}5 \cdot \frac{1}{2}}} = \underline{0{,}0235}$$

c) $$E[T] = \int_{-\infty}^{\infty} t \cdot f(t)\, dt = \int_{0}^{\infty} t \cdot 1{,}5 e^{-1{,}5t}\, dt$$

$$= \int u' v\, dx = u \cdot v - \int u v'\, dx$$

$u' = 1{,}5e^{-1{,}5t} \qquad u = -e^{-1{,}5t}$

$v = t \qquad v' = 1$

$$-te^{-1{,}5t} + \int e^{-1{,}5t}\, dt \Big|_0^{\infty}$$

$$-te^{-1{,}5t} + \left(-\frac{1}{1{,}5} e^{-1{,}5t}\right) \Big|_0^{\infty}$$

$e^{-1{,}5t}$ vokser raskere mot 0 enn

$t \to \infty$

$\Rightarrow$

$$(0+0) - \left(0 - \frac{1}{1,5}\right) = \frac{1}{1,5} = \frac{2}{3}$$

5) $P(x=0 \mid y=1) = \dfrac{P(x=0 \wedge y=1)}{P(y=1)}$

$$= \frac{0,35}{0,45} = 0,\overline{77}$$

$$E[Y] = \sum_{y} y \cdot f(y) = \underbrace{0 \cdot 0,1 + 0 \cdot 0,35}_{0,35+0,1} \quad (-1)$$

$$+2 \cdot 0,1 + 2 \cdot 0,05$$

$$= 0,75$$

$$E[XY] = \sum_{xy} (xy) \cdot f(x,y)$$

$x \cdot y$	$f(x,y)$
$1 \cdot 1$	$0,1$
$1 \cdot 2$	$0,05$

(ignorer $x \vee y = 0$)

$$E[XY] = 1 \cdot 0,1 + 2 \cdot 0,05 = \underline{\underline{0,2}}$$

6)

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1 - \cos(2x)}{2} & 0 \le x \le \frac{\pi}{2} \\ 1 & x > \frac{\pi}{2} \end{cases}$$

X = kontinuerlig

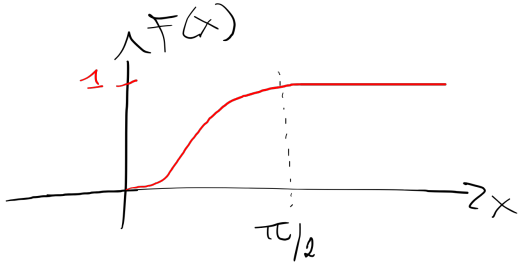
a) $F'(x) = f(x) \quad x > 0$

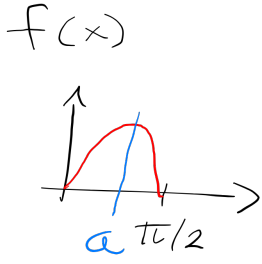
$0 \le x \le \frac{\pi}{2}$: $F'(x) = \frac{d}{dx}\left(\frac{1}{2} - \frac{\cos(2x)}{2}\right)$

$$= -\frac{1}{2} \cdot 2 \cdot (-\sin(2x)) = \sin(2x)$$

$x > \frac{\pi}{2}$: 0

b) $E[X] = f(x)\Big|_{-\infty}^{\infty}$





$E[X] = a \iff f(a-z) = f(a+z)$

$z \in \mathbb{R}$

$\Rightarrow f'(a) = 0$

$\frac{d}{dx}\sin(2x) = 0 \Rightarrow -2\cos(2x) = 0$

$\cos(2x) = 0$

$2x = \frac{\pi}{2} \quad \left| \; 0 \leq x \leq \frac{\pi}{2} \right.$

$a = \frac{\pi}{4}$

c) $E[X^2] = Var[X] + (E[X])^2$

$(X)^2$ (broke sense i d)

$$E[X^2] = \int x^2 \cdot f(x)\,dx = \int x^2 \sin(2x)\,dx$$

$$x^2 \int \sin(2x)\,dx - \int 2x \cdot \int \sin(2x)\,dx\,dx$$

$\int \sin 2x\,dx = -\frac{1}{2}\cos(2x)$ (sin → cos)

$$x^2 \cdot \frac{-1}{2}\cos(2x) + \int 2x \cdot \frac{1}{2}\cos(2x)\,dx$$

$$-\frac{1}{2}x^2\cos(2x) + \int x\cos(2x)\,dx$$

$$= -\frac{1}{2}x^2\cos(2x) + x\int\cos(2x)\,dx - \int\int\cos(2x)\,dx\,dx$$

$$= -\frac{1}{2}x^2\cos(2x) + \frac{1}{2}x\sin(2x) - \int\frac{1}{2}\sin(2x)\,dx$$

$$= -\frac{1}{2}x^2\cos(2x) + \frac{1}{2}x\sin(2x) - \frac{1}{2}\left(-\frac{1}{2}\right)\cos(2x)$$

$$\int_0^{\pi/2} = [\quad]_0^{\pi/2}$$

$= 0,773\ldots$

d) $Var[X] = E[X^2] - (E[X])^2$

$= 0,733 - \left(\frac{\pi}{2}\right)^2 = 0,1168$

$Var[X] = E\left[(X-\mu)^2\right] \quad \mu = E[X]$

$= \int_0^{\pi/2} \left(x - \frac{\pi}{4}\right)^2 \sin(2x)\, dx$

e) $SD[X] = \sqrt{Var[X]}$